

School Ratings, Language Barriers, and Enrollment Choices of Limited English Proficiency Families: Evidence from Texas Elementary Schools

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Abstract

Public school ratings are intended to reduce the costs of acquiring and interpreting information about school quality, but their effectiveness depends on the ability of families to access and act on the information provided. I study whether school accountability ratings affect the enrollment choices of Limited English Proficient (LEP) students, who may face greater language barriers when making school-choice decisions. I combine school accountability records with restricted student-level administrative data from Texas elementary schools and exploit statutory performance thresholds that sharply increase the probability that a school receives a higher rating. I find little evidence of an enrollment response among Hispanic LEP students at any of the three rating thresholds. At the highest rating threshold, by contrast, enrollment of non-LEP students increases by 1.9 percent and incoming transfers increase by 1.5 percentage points. More demographically comparable non-LEP groups exhibit positive but less precise responses at the lowest rating threshold, and the estimated differences between Hispanic LEP and non-LEP students are not statistically distinguishable from zero. Heterogeneity estimates suggest a larger Hispanic LEP response in schools with high LEP concentration. These findings show that making school-quality information publicly available does not necessarily generate uniform behavioral responses, particularly among families facing multiple information and choice frictions.

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1 Introduction

State governments across the US routinely publish school accountability reports to make school quality information easily accessible for families. But accessing, interpreting, and evaluating school quality based on these reports might be cumbersome and costly for most parents (Hastings and Weinstein (2007); Agte et al. (2024)). Accountability ratings compress this information into a simple and salient signal and are therefore designed to reduce the costs of acquiring and interpreting information about school quality. Yet the effectiveness of such transparency policies depends on an implicit assumption that families are able to access, understand, and act on the information being provided.

Families differ substantially in the costs of acquiring and using school-quality information. These costs may be particularly high for families of Limited English Proficient students, who can face language barriers in addition to the typical search costs faced by other households when choosing a school. Since LEP status of a family could also be correlated with other factors such as socioeconomic disadvantage, they can be considered representative of a population facing multiple potential information and choice frictions rather than as purely a measure of language barriers.

These frictions create an important tension for school accountability policy. A simple rating could be especially valuable for families facing high information costs because it provides an alternative for more costly information acquisition. On the other hand, if those same barriers prevent families from learning about or interpreting the rating itself, accountability policies may disproportionately benefit households that already face relatively low information frictions. Therefore, whether school ratings reduce or reinforce disparities in responsiveness is ultimately an empirical question.

This paper asks whether school accountability ratings affect school choice decisions by families facing high costs of acquiring school quality information as evidenced by their enrollment choices. I study Texas elementary schools between 2001 and 2008, when the state’s accountability system assigned schools to one of four categorical ratings based on predetermined performance thresholds. This setting is useful for several reasons. First, unlike much of the existing evidence on information and school choice, students in this

setting were primarily assigned to schools based on residence rather than being required to rank schools through a centralized choice mechanism. Families could nevertheless respond to information about school quality through residential mobility or transfers. Second, Texas was one of the earliest adopters of the accountability system dating back to early 1990s. So, during the period of the study, parents have had considerable exposure to the formal rating system. Finally, the accountability formula used by Texas during the study period generates discrete changes in a school rating around continuous measures of underlying performance.

I exploit these rating thresholds in a regression discontinuity design to isolate the causal effect of receiving a higher accountability rating. Schools immediately above and below a threshold have similar underlying performance but receive different categorical signals about school quality. So, comparing enrollment outcomes around these cutoffs allows me to distinguish responses to the public rating from responses to the underlying characteristics that determine school performance. It is important to note that the design only examines whether the categorical label itself changes behavior among schools with very similar underlying performance. It does not test whether families respond to larger differences in school quality or to ratings when schools are farther apart in the underlying distribution. I then compare the effects of LEP students with non-LEP students and test whether their enrollment responses differ. This comparison allows me to benchmark the LEP estimates and provides evidence of heterogeneity by LEP status.

The raw enrollment patterns are highly correlated with accountability ratings. The gap in proportional enrollment growth between Hispanic LEP, which make up a majority of LEP students, and Hispanic non-LEP students rises monotonically with school quality, from -11 percentage points in schools rated Unacceptable, which was the lowest rating tier in Texas during the study period, to +16 percentage points in schools rated Exemplary, which was the highest rating tier. Taken at face value, these estimates suggest that Hispanic LEP families were sorting sharply towards better-rated schools. However, schools receiving different ratings also differ systematically in several factors such as student composition, neighborhood characteristics, socioeconomic status, and the underlying quality and availability of schooling options. So, simple comparisons across rating

categories therefore cannot isolate the informational effect of the rating itself.

But when I use the regression discontinuity design, the differences in response disappear. I find little evidence of a discontinuous response among Hispanic LEP students at any of the three rating thresholds. This contrasts with the positive and relatively robust increase in enrollment and incoming transfers of the broader non-LEP student population at the Exemplary threshold. However, among Hispanic non-LEP and economically disadvantaged non-LEP students, who are a more demographically comparable benchmark to Hispanic LEP students, the estimates are also positive for enrollment growth and incoming transfers at the Acceptable threshold but are imprecise and sensitive to bandwidth choice. At the other rating thresholds, these groups also show little evidence of enrollment response. Taken together, the results show a weak response to school rating labels among Hispanic LEP students while the evidence of a differential response among other groups of non-LEP students is more suggestive than conclusive.

Even though the ratings and the underlying pass rates that determine the rating are public, the question remains why the Hispanic LEP families show little response to crossing the rating threshold. There could at least be three possible reasons. First, the ratings may not reach or be salient to these families. Second, families may be aware of the ratings but face constraints that limit their ability to act on it. These could be financial constraints or limited choice of schools around them. Third, families may already be sophisticated in their choices and understand that the schools around the rating threshold are of similar quality and therefore rationally place little weight to the label itself. This study is unable to distinguish among these channels but explores whether the weak average response masks meaningful heterogeneity across school environments.

I first examine whether LEP families are more responsive when they have more schooling choice within their district. There seems to be little evidence that Hispanic LEP students respond more strongly in districts with above-median number of schools. Second, I test their responsiveness when they are surrounded by a larger population of students facing similar language-related schooling needs. In schools where above-median Hispanic LEP enrollment share at baseline, the Hispanic LEP enrollment significantly increases by 7.2 percentage points on average at the Acceptable threshold. This shows

that the overall null does not reflect a complete lack of responsiveness to accountability ratings among Hispanic LEP households but appears to vary with the environment in which families encounter and act on the rating.

This paper contributes to several growing strands of literature. First, there are number of studies that show that accountability systems have real effects on schools and educators. For example, Neal and Schanzenbach (2010) found that introduction of test-based accountability systems generated increased test scores for students in the middle of the academic distribution and not for the least academically advantaged students. Meanwhile, Dee and Jacob (2011) showed that similar systems led to increases in test performances across the distribution in some subjects but not in others. Beyond the academic margin, Cullen et al. (2024) found that school ratings have affect the career outcomes of school principals. These studies establish that school accountability ratings are consequential signals and are not simply labels.

A related literature studies whether families respond to school quality information. Figlio and Lucas (2004) show that state-assigned school grades affect housing prices and residential sorting. Similarly, experimental evidence shows that lowering the cost of acquiring school quality information can affect schooling choices made by households. Hastings and Weinstein (2007) found that providing low-income households with simplified information on school test scores shifted their choices towards high-performing schools. In a different setting, Andrabi et al. (2017) randomly provided school report cards to parents in Pakistani schools and found an improvement in test scores and enrollment outcomes. My paper complements this body of research by studying the impact of information that is less salient to parents by being publicly released as a part of the existing accountability system rather than being delivered to them through a direct intervention. This distinction is crucial to study because making information available does not ensure that families receive, process, or use it.

Some recent work has emphasized that the effectiveness of school quality information depends on the information and search frictions that families face. Cohodes et al. (2025) employed large-scale informational intervention in New York City high schools and found that the mode and simplicity of information delivery matter in student school choice

responses. Most relevant for this study was the finding that English learners were the strongest responders to all the information interventions. Agte et al. (2024) is another useful study for my setting as they show that families may be unaware of the full set of schools available to them and may also misperceive the quality of schools that they do know. These findings suggest that the existence of a simple public signal does not imply that all households can make equal use of it, which is pertinent to my study as I focus on a population that is likely to face relatively high information and navigation costs.

Finally, this study extends the literature on students classified as English learners. Most of the research work until now has examined the policies that directly affect the education environment of these student, including provision of bilingual education (Chin et al. (2013)) or the reclassification out of English-learner status (Carlson and Knowles (2016)). I shift the focus from curriculum to how families of LEP students use general information about overall school quality.

Broadly, this paper takes the literature forward by studying who responds to a simplified public signal of school quality in a context where acquiring and acting on information may be costly. Unlike much of the literature that uses information-related interventions, the setting in this study does not include a centralized choice process or direct provision of information to families. Instead, the research design isolates the local effect of a categorical rating in a neighborhood-based system of schooling and examines how the response varies by LEP status and across different school environments.

2 Institutional Setting and Data

2.1 The Texas School Accountability System

Texas was one of the earliest adopters of a school accountability system. By the early 2000s, which is the period of focus in this study, the Texas Education Agency (TEA) annually assigned each public school a categorical rating that was primarily based on student performance on state-wide standardized exams. These ratings were explicitly made public and every school was mandated to provide a copy of the accountability report to parents. An example of the information provided in these reports is shown in Figure A1. The school accountability rating on top of the report card is a relatively

coarse but salient summary of the school performance in contrast to the test score performance reported along with it. The ratings also received widespread media coverage and publication alongside real estate listings.

In this study, I focus on Texas elementary school from 2000-01 through 2007-08 academic years. For convenience, I will refer to an academic year by its ending year. I exclude 2003 from the analysis as Texas was transitioning to a different testing system and no ratings were published in that year. In all remaining years, schools were classified into one of four rating categories: Unacceptable, Acceptable, Recognized, or Exemplary.

The rating was determined using pass rates on standardized tests for multiple subjects. The tested subjects included mathematics and reading in grades 3 through 6, writing in grade 4, and science in grade 5. Each school's performance is evaluated separately for all students, Hispanic students, White students, Black students, and economically-disadvantaged students. A group except all students was only included if they exceeded the minimum size requirement, which was typically 30 students. The pass rates of each eligible subject-by-group was compared with statutory performance thresholds that varied by accountability level and across years. The threshold for exemplary rating was typically a pass rate of 90 percent; the recognized threshold ranged from 70 to 80 percent; while the acceptable threshold was lower and varied more substantially across subjects and years.

The accountability system was designed to incentivize schools to focus on all subjects and students - especially the low-performing groups. It effectively imposed a minimum-performance rule where a school's overall rating could be no higher than the performance of its lowest-performing subject-by-group. I refer to the eligible subject-by-group cell with the lowest pass rate as the school's *marginal group*. This feature of the system is central to the empirical design because it allows school performance to be summarized relative to the statutory cutoff relevant for a particular accountability margin.

The system also included additional provisions to reward improvement and address circumstances perceived to unfairly disadvantage some schools. The rules allowed schools that fell short of a performance threshold to qualify for a higher rating under *required improvement* provisions when performance had improved sufficiently relative to the prior

year. Further, recognizing that more subgroups increase the probability of a low rating for a given overall distribution of student knowledge, schools could also receive a limited number of *exceptions* when particular measures narrowly failed to satisfy the relevant standard. There were restrictions to use of required improvements and exceptions. For example, a subgroup could not be exempted two years in a row.

I incorporate these provisions when reconstructing each school’s accountability rating. For every school-year, I calculate performance for each eligible subject-by-student group, apply the contemporaneous statutory thresholds, required-improvement provisions, and exception rules, and assign the final rating implied by the full accountability formula. I then identify the marginal group and measure its performance relative to the statutory cutoff associated with each rating margin. Thus, separate running variables can be constructed for the Unacceptable-to-Acceptable, Acceptable-to-Recognized, and Recognized-to-Exemplary margins.

These rules generate the identifying variation used in the paper. Schools with marginal-group performance just above and just below a statutory cutoff have very similar underlying test performance but face a discontinuous change in their likelihood of receiving the higher accountability rating. Because required-improvement and exception provisions can affect the final rating, crossing the statutory threshold does not mechanically determine treatment in every case.

2.2 Data and Sample

I combine two primary sources of data for this study. First, I use the Texas Academic Excellence Indicator System (AEIS), which contains indicators on school accountability ratings and the pass rates used to calculate them. I reconstruct the school ratings and identify the demographic group that determines the relevant rating threshold for each school in each year based on the rating process followed in Cullen et al. (2024). Second, I use the restricted student-level administrative records from the Texas Education Agency (TEA) provided by the Education Research Center (ERC) at the University of Texas at Dallas Texas Schools Project. The data allows me to follow students longitudinally across schools in Texas and contains information on their enrollment, academic performance

and demographic characteristics. I use this data to create the outcome indicators on student mobility across schools.

The focus of the study is on Limited English Proficient (LEP) students. During the period I study, Texas identified LEP students through state-approved oral language proficiency tests. I classify a student as LEP if they are observed as Limited English Proficient at any point in the dataset during the sample period. A student who is never observed with a LEP classification is categorized as non-LEP. LEP students account for 29.2 percent of the total enrollment in my sample. Among LEP students, 91.5 percent are Hispanic and 94.6 percent are ever eligible for free or reduced-price lunch. In comparison, only 31 percent of non-LEP students are Hispanic and 63.7 percent of them are ever eligible for free or reduced-price lunch. I refer to the latter measure as *ever economically disadvantaged* henceforth in this paper.

Since LEP students in Texas are predominantly Hispanic, my main analysis will focus on them and will compare their outcomes to Hispanic non-LEP students. Restricting the comparison to Hispanic students is deliberate to hold race and ethnicity fixed. However, since Hispanic non-LEP is a smaller sample, I also report estimates using all non-LEP and economically disadvantaged non-LEP students as alternative comparison groups.

Table 1 reports the characteristics of Hispanic LEP and Hispanic non-LEP students in the analysis sample. Panel A shows that 97.4 percent of Hispanic LEP students are economically disadvantaged, which is substantially more than 84.8 percent for Hispanic non-LEP students. Hispanic LEP students are also slightly more concentrated in urban areas with 61.4 percent attending urban schools compared to 56.8 percent of Hispanic non-LEP students. Finally, the average school enrolls about 155 Hispanic LEP students and 128 Hispanic non-LEP students.

Panel B shows the distribution of the two groups across accountability categories. Among Hispanic LEP students, more than half are enrolled in schools rated Acceptable while only 7.3 percent attended school rated Exemplary. In comparison, among Hispanic non-LEP students, the corresponding figures are 41.3 percent and 11.6 percent respectively. These descriptive differences show that LEP status of students is correlated with school-level characteristics and their exposure to accountability ratings. So, simply comparing

enrollment behavior across rating categories may reflect underlying differences in student and school characteristics rather than causal effect of the rating itself. This motivates the regression discontinuity design outlined in Section 3.

2.3 Measuring Enrollment Responses

The outcomes of interest in this study are designed to capture the changes in revealed preference for a school following the publication of its accountability rating. Since the focus is on change, it is important to be careful about timing in this setting. A rating is published based on tests administered in academic year t in the early months of the academic year $t + 1$. Since enrollment decisions are already made by the time ratings are published, I measure enrollment outcomes in year $t + 2$. This allows parents sufficient time to observe and respond to the rating.

The primary outcome for the study is enrollment change between year t and $t + 2$. This measure captures the net change in group-specific enrollment between the rating year and the post-rating enrollment year. The remaining outcomes distinguish the different channels of enrollment change. While isolating the channels, it is important to distinguish between incumbent and prospective families. Families whose children are already enrolled in a school will observe many aspects of quality directly and therefore might not see accountability ratings as information shocks. So, families considering entry into a school might place a higher weight on public ratings. For this reason, while the study investigates both entry and exit margins, the former is particularly interesting.

An important complication is that many school changes are structural in nature. For example, students moving into a feeder school after graduating from a terminal grade should not be seen as a response to a school rating. Therefore, I only focus on *nonstructural* moves in the entry and exit outcome measures. Additionally, I restrict the sample to schools that offer the same set of grades between year $t - 2$ and the outcome year $t + 2$. This ensures that changes in the school's grade configuration does not mechanically generate large changes in enrollment when examining balance at baseline or the effect on outcome variables.

I separate entry into *incoming transfers* and *new entrants* to distinguish families

based on their possible level of familiarity with the Texas school system. To identify incoming transfers, I follow a student observed in year t from their school to another school in year $t + 2$. I classify an incoming transfer as a structural transition when more than 35 percent of the origin cohort moves to the same destination. I exclude these flows and define incoming transfer as students arriving at a focal school through remaining nonstructural flows. This helps me isolate lateral student movement that is more likely an active schooling choice. However, it could be the case that structural moves are also active choices when there are more than one feeder school options available. So, the estimates likely understate the incoming transfer response. I separately define new entrants as students enrolled at the focal school in $t + 2$ who were not observed in the Texas public school system in t . These families could be either enrolling their children at pre-kindergarten level or entering the state for the first time.

Finally, I define *leavers* as students who were enrolled at the focal school in t but are no longer enrolled there in $t + 2$. In this case, I exclude structural leavers by excluding students who are in the school's terminal grade or one year below the terminal grade in t since the outcome horizon spans over two years and both groups are expected to graduate out of the school by $t + 2$.

I normalize all outcome variables by the group's enrollment at the school in the baseline year. So, the resulting estimates can be interpreted as changes relative to the baseline group enrollment.

2.4 Descriptive Enrollment Patterns

Figure 1 presents raw mobility patterns by accountability rating for all outcome variables. The enrollment growth pattern varies sharply by rating category and also differs for Hispanic LEP and Hispanic non-LEP students. The same holds true on the entry margin as well, especially for incoming transfers. However, these differences in mobility cannot be attributed to school rating labels since there are systematic differences in the schools and the types of students enrolled in these schools. Therefore, I turn to a regression discontinuity design to isolate the true effect of the rating.

3 Empirical Strategy

3.1 Identification Strategy

The primary challenge in understanding the effect of school rating on enrollment outcomes of students is that schools of different rating categories also differ in other meaningful factors such as neighborhood characteristics, quality of teachers and peers, parental resources and so on. A comparison of enrollment outcomes across rating categories will therefore confound true responses to accountability ratings with these underlying differences.

I address this challenge by using a regression discontinuity (RD) design based on the unique accountability framework followed by Texas during the sample period. As described in Section 2.1, a school’s rating was typically determined by the performance of the lowest performing subject-by-demographic group. In this setting, schools whose marginal group pass rates fall just above and below a threshold are of similar underlying quality but face a discrete change in the probability of receiving a higher accountability rating. The resulting discontinuity can be exploited to isolate the variation in the public signal of school quality.

I construct a separate running variable based on the pass rate of the marginal group for each of the three rating thresholds: Unacceptable-Acceptable, Acceptable-Recognized, and Recognized-Exemplary. I will refer to these as the Acceptable, Recognized, and Exemplary thresholds.

Let p_{st} denote the pass rate of the marginal group for school s in year t , and let c_{st}^k denote the statutory threshold that applies to that group at rating margin $k \in \{A, R, E\}$. I define

$$R_{st}^k = p_{st} - c_{st}^k. \quad (1)$$

The running variable is centered at zero. So, schools with $R_{st}^k < 0$ fall below the statutory threshold and schools with $R_{st}^k \geq 0$ meet or exceed it. The centered running variable is necessary because the applicable thresholds differ across years.

Each school’s rating and marginal group is reconstructed using the full accountability

assignment rules based on Cullen et al. (2024), including the statutory pass-rate thresholds, required-improvement provisions, and exceptions. The reconstructed assignment rule predicts approximately 98 percent of observed school ratings. Since crossing the statutory threshold changes the probability of receiving the higher rating but does not perfectly determine the observed rating, I treat the design as a fuzzy regression discontinuity. Consequently, all reported estimates are intent-to-treat effects of crossing the threshold.

3.2 Estimation

For each rating margin, I estimate the following local-linear reduced-form specification:

$$Y_{sdt} = \alpha + \beta \mathbb{1}\{R_{sdt}^k \geq 0\} + \gamma_1 R_{sdt}^k + \gamma_2 \mathbb{1}\{R_{sdt}^k \geq 0\} \times R_{sdt}^k + \lambda_{dt} + \varepsilon_{sdt}, \quad (2)$$

where Y_{sdt} is an enrollment or mobility outcome for school s in district d and year t , R_{sdt}^k is the centered running variable for rating margin k , and λ_{dt} denotes district-by-year fixed effects. The specification allows the relationship between the outcome and the running variable to differ on either side of the threshold.

The coefficient of interest, β , measures the discontinuous change in the outcome when the school’s marginal-group pass rate crosses the statutory rating threshold. I estimate Equation 2 separately at the Acceptable, Recognized, and Exemplary margins and separately for Hispanic LEP and Hispanic non-LEP students. The baseline specification uses observations within eight percentage points of the relevant cutoff and clusters standard errors at the campus level. Because the outcomes are normalized by the corresponding group’s year- t enrollment, the coefficients can be interpreted as changes in the enrollment-flow ratio associated with crossing the rating threshold.

The corresponding first-stage specification is:

$$D_{sdt}^k = \pi_0 + \pi_1 \mathbb{1}\{R_{sdt}^k \geq 0\} + \pi_2 R_{sdt}^k + \pi_3 \mathbb{1}\{R_{sdt}^k \geq 0\} \times R_{sdt}^k + \lambda_{dt} + u_{sdt}, \quad (3)$$

where D_{sdt}^k indicates whether the school receives a rating at or above the relevant category. The coefficient π_1 measures the discontinuous change in the probability of receiving the higher rating at the statutory cutoff.

Figure 2 presents the first stage for each of the three rating margins. At all three thresholds, crossing the statutory cutoff produces a large increase in the probability that a school receives the higher accountability rating. The discontinuity is particularly sharp at the Recognized and Exemplary thresholds, where crossing the cutoff increases the probability of receiving the higher rating by approximately 98.0 and 95.3 percentage points, respectively. The first stage at the Acceptable threshold is also large but the assignment is somewhat less sharp at about 88.3 percentage points. This imperfect compliance stems from the limitations in perfectly replicating the rating process due to institutional complexities, which allow some schools below the statutory threshold to receive a higher final rating and are unobserved by the researcher. Overall, the plots confirm that the statutory thresholds generate strong discontinuities in the public rating received by schools.

3.3 Identification and Validity

The validity of this RD design depends on the assumption that in the absence of the discrete change in the accountability rating, potential enrollment outcomes vary smoothly through the rating threshold (Lee and Lemieux, 2010; Cattaneo and Titiunik, 2022). This condition would be violated if schools could precisely manipulate their position relative to the cutoff or if other determinants of student mobility changed discontinuously at the threshold.

A key evidence against precise manipulation is that schools should not be able to cluster above the cutoff (McCrary, 2008). Figure 3 plots the distribution of the running variable around each of the three rating thresholds. I do not observe an obvious discontinuity or bunching above zero at any of the cutoffs. While the distribution varies over the wider support of the running variable, its local behavior around the threshold provides no clear visual evidence of precise sorting. Further, given the complex nature of the rating mechanism, it is difficult for any school to precisely manipulate their position around the cutoffs.

As a second assessment of the identifying assumption, I test whether predetermined enrollment characteristics and pre-period mobility outcomes vary discontinuously at

the rating thresholds. Tables 2 and 3 report these balance tests for Hispanic LEP and Hispanic non-LEP students, respectively. Each entry reports the estimated discontinuity at the relevant cutoff within a bandwidth of 8 points and the mean of the corresponding variable is reported below.

Table 2 shows that predetermined characteristics are generally smooth through the thresholds for Hispanic LEP students. Neither total LEP enrollment nor Hispanic LEP enrollment exhibits a significant discontinuity at any of the three thresholds. Similarly, pre-period enrollment growth, incoming transfers, and new entrants show no significant jumps. The only exception is the pre-period leaving ratio at the Exemplary threshold, which increases by approximately 3 percentage points at the cutoff. Therefore, the balance tests are broadly supportive of local comparability for the LEP sample. However, the pre-period discontinuity in leavers at the Exemplary margin warrants some caution when interpreting estimates at that threshold.

The balance is somewhat less clean for the non-LEP sample in Table 3. At the Acceptable threshold, the pre-period incoming transfer ratio increases by approximately 6 percentage points and is statistically significant. At the Recognized threshold, total non-LEP enrollment exhibits a marginally significant discontinuity, which is reflected in the enrollment change ratio as well, while Hispanic non-LEP enrollment and the pre-period mobility measures remain statistically smooth. The evidence of imbalance at the Acceptable threshold is not robust to a narrower bandwidth. Appendix Table A1 presents the balance tests using a bandwidth of four percentage points. The estimated discontinuity around the Acceptable threshold for incoming transfers falls to zero is no longer statistically significant. However, the non-LEP enrollment discontinuity around the Recognized threshold seems persistent. Consequently, I interpret the non-LEP estimates along this margin more cautiously and examine their sensitivity to alternative bandwidths and specifications.

4 Main Results

4.1 Enrollment Responses to Accountability Ratings

The reduced form effects of crossing each accountability threshold on enrollment change are shown in Table 4. The first column presents estimates for Hispanic LEP students while the remaining columns report estimates for all non-LEP students, Hispanic non-LEP students, and economically disadvantaged non-LEP students. The latter groups provide comparable benchmarks along dimensions that are strongly correlated with LEP status. All non-LEP student groups primarily include Hispanic and non-Hispanic white students. The Hispanic non-LEP student group is more comparable to the Hispanic LEP group in their background and language-based networks. But since the Hispanic LEP group is predominantly low-income as shown in Table 1, the economically disadvantaged non-LEP group is comparable along that margin.

For Hispanic LEP students, there appears to be little evidence of an enrollment response at any of the three rating thresholds. Crossing the Acceptable threshold increases Hispanic LEP enrollment by 2.7 percent relative to baseline enrollment, but the estimate is imprecise and statistically indistinguishable from zero. The corresponding estimates at the Recognized and Exemplary thresholds are -1.9 and -3.1 percent, respectively, and are also statistically insignificant. By contrast, some of the non-LEP comparison groups exhibit larger positive responses at particular thresholds. At the Acceptable threshold, the enrollment for all non-LEP students increases by 5.4 percent and for economically disadvantaged non-LEP students increases by 4.3 percent. Both of the estimates are higher in magnitude but also statistically indistinguishable from zero. On the other hand, the Hispanic non-LEP enrollment grows by 16.5 percent and is statistically significant. The large magnitude should be interpreted with caution given the small sample of schools around the Acceptable threshold and Hispanic non-LEP being a smaller student group on average.

Figure 5 provides the graphical counterpart to these estimates for Hispanic LEP and all non-LEP students. The top row shows little evidence of a visible discontinuity in Hispanic LEP enrollment at any of the three thresholds. The bottom row shows a different pattern

for all non-LEP students. In particular, there is a visible upward shift in enrollment at the Exemplary cutoff, which is consistent with the positive estimate reported in Table 4. The graphical evidence around the Acceptable threshold is comparatively noisier while there is little indication of a meaningful discontinuity at the Recognized threshold for either group.

Appendix Table A2 examines the sensitivity of these estimates to alternate bandwidths. The coefficients are more noisy around the Acceptable threshold for all student groups including Hispanic LEP, which is consistent with the smaller number of observations around this threshold. The pattern at the Exemplary threshold is notably more stable by contrast. For all non-LEP students, the estimated enrollment response remains positive and very similar in magnitude across all bandwidths. By comparison, the Hispanic LEP coefficient remains negative and imprecise across all of the bandwidth specifications including the optimal bandwidth.

The key takeaway is that the enrollment response to rating is more muted for Hispanic LEP students relative to some of the comparison groups, particularly at the Exemplary threshold. Even at the Exemplary threshold where the enrollment for all non-LEP students increases by 1.9 percent in the baseline specification, the estimate for Hispanic LEP students is negative and not statistically different from zero.

While these estimates and the graphical evidence suggest that Hispanic LEP students may respond less strongly to accountability ratings than some non-LEP students, they do not themselves imply that the underlying responses are statistically different. I therefore estimate the LEP–non-LEP differences directly. Table 5 reports these comparisons. First, almost all of the differences between Hispanic LEP and the comparison groups are negative underlying the fact that the Hispanic LEP response is much smaller. Second, the differences are substantially smaller across most thresholds in comparison with economically disadvantaged non-LEP students. This could be indicative of the level of similarity between the two groups. The similarity could exist along different margins such as awareness and understanding of the rating or the level of resources. Finally, none of the differences are statistically distinguishable from zero, which includes the difference with Hispanic non-LEP on the acceptable margin and all non-LEP on the exemplary

margin.

Taken together, these estimates do not provide statistically precise evidence that Hispanic LEP students respond differently to accountability ratings than non-LEP students. The point estimates, however, are consistent with a more muted response among Hispanic LEP students relative to the broader non-LEP population at some rating margins, particularly the Exemplary threshold. At the same time, the differences are noticeably smaller when the comparison group is restricted to economically disadvantaged non-LEP students. This suggests that part of the contrast between Hispanic LEP and all non-LEP students may reflect differences in socioeconomic composition rather than LEP status alone.

4.2 Enrollment Channels

I next examine whether the enrollment responses operate through entry into the Texas public school system, or students leaving the school. Table 6 reports the RD estimates for each of these three margins. I treat these outcomes as secondary to the overall enrollment change measure. Enrollment change captures the net effect of student mobility, whereas the channel outcomes measure gross flows and are therefore more sensitive to differences in the overall level of student turnover across schools.

The clearest pattern appears for incoming transfers. At the Acceptable threshold, crossing the cutoff increases the incoming-transfer ratio by 3.3 percent for the Hispanic LEP students but the estimate remains insignificant. By contrast, it increases by 9.5 percentage points for all non-LEP students, 17.7 percentage points for Hispanic non-LEP students, and 10.3 percentage points for economically disadvantaged non-LEP students. All three estimates are positive, and the estimates for all non-LEP and economically disadvantaged non-LEP students are statistically significant at conventional levels. However, these results must be interpreted with caution as the incoming transfers for all non-LEP students was imbalanced at baseline on the Acceptable threshold. At the Exemplary threshold, which was more balanced in the pre-period, incoming transfers also increase by 1.5 percentage points for all non-LEP students while the Hispanic LEP estimate is negative and imprecise. There is little evidence of an incoming transfer

response at the Recognized threshold.

Appendix Table A3 shows the robustness of these effects to bandwidth selection. The Hispanic LEP coefficients are imprecisely estimated on all thresholds across all bandwidth specifications. The incoming transfers for all non-LEP students around the Exemplary threshold seems robust to all bandwidths except below 6. However, the magnitude of the coefficient appears similar even when the bandwidth is 4. The discontinuity also seems robust at the Acceptable threshold for all non-LEP and economically disadvantaged non-LEP students. However, it is difficult to put too much weight on the effects on this threshold due to issues of sample size and pre-period imbalance.

The remaining mobility margins show considerably weaker responses. I find no statistically significant effect on new entrants for Hispanic LEP students or for the broader non-LEP comparison groups at any threshold. Similarly, the leaver estimates are generally small and statistically indistinguishable from zero. Thus, the non-LEP enrollment response to the rating comes strongly from students transferring schools but Hispanic LEP students remain unresponsive even on that front.

Overall, the main results reveal that Hispanic LEP enrollment does not exhibit a notable response in enrollment to crossing the accountability thresholds, while some non-LEP groups respond positively on some margins through incoming transfers between schools.

The absence of a precisely estimated enrollment response among Hispanic LEP students does not tell us why these families appear less responsive to accountability ratings. Several mechanisms could generate this pattern. First, families may face information frictions that limit whether they receive the rating signal. Second, families may observe the rating but face constraints that limit their ability to respond by changing schools. These constraints could be financial as it is costly to move neighborhoods for schooling or institutional as there might be limited schools in proximity, especially ones that cater to needs of LEP families. Third, families may recognize that schools immediately above and below a rating threshold are similar in underlying performance and therefore place relatively little weight on the categorical label itself.

The available data does not allow me to distinguish cleanly among these mechanisms.

Instead, I examine whether responsiveness varies across school environments in ways that are informative about some of these possibilities. I first consider the local choice environment. If limited schooling options constrain families' ability to respond to new information, ratings should have greater scope to affect enrollment in districts with multiple elementary schools than in districts where families have few within-district alternatives. I then examine heterogeneity by the concentration of LEP students in the school. Schools with a high share of LEP students may differ in the availability of language-related programs, and the presence of Spanish-speaking families. These factors could either facilitate information transmission or reduce families' willingness to move in response to accountability ratings. The resulting estimates should therefore be interpreted as evidence on how responsiveness varies across school environments rather than as tests of a single mechanism.

5 Heterogeneity Results

I examine whether the enrollment response varies across two dimensions of the school environment: the local choice set and the concentration of LEP students. I define a high-choice district as one where the number of elementary schools in the district is above median and low-choice districts have below-median number of elementary schools. Similarly, I define a high-LEP school as one with an above-median share of LEP students, while low-LEP schools have an LEP share that is below the median.

The idea behind the first split is to capture differences in the local schooling options available to families. Ideally, this would be measured using the number of elementary schools located close to a family's place of residence. But I do not observe school district boundaries or precise residential choice sets in my data. So, I use the number of elementary schools in the administrative district as a proxy for the set of nearby schooling options available to families.

The second split captures differences in the concentration of LEP students within the school. A higher share of LEP students may proxy for the presence of more students and families from similar linguistic backgrounds, which could make a school more attractive to LEP families. It may also be correlated with school-level characteristics

that are particularly relevant for LEP students such as the availability of language-related programs. Therefore, I interpret LEP concentration as a broad measure of the school environment faced by LEP families rather than as a direct measure of any single mechanism. I remain agnostic on their preference for high or low LEP concentration.

Before presenting these heterogeneity estimates, Table 7 describes how the high- and low-choice and high- and low-LEP school environments differ along observable dimensions. Panel A compares high- and low-choice districts. The two groups differ substantially in their observable characteristics. Low-choice districts have one elementary school on average while high-choice districts have approximately 43 elementary schools. High-choice districts are also substantially more urban. More than half of schools in high-choice districts are urban compared with only 4.2 percent in low-choice districts. Schools in high-choice districts are also larger and have a higher share of LEP students. LEP students account for 28.8 percent of enrollment in high-choice districts compared with 10.6 percent in low-choice districts. The share of economically disadvantaged is almost similar across both groups. These patterns suggest that the choice measure captures not only the number of available schools but also broader differences between large urban and smaller rural districts.

Panel B shows that high-LEP schools also differ considerably from low-LEP schools. By construction, the LEP share is substantially higher, at 45.3 percent compared with 7.2 percent in low-LEP schools. High-LEP schools are also larger, more urban, and more economically disadvantaged. They are located in districts with more elementary schools as well. The average high-LEP school is in a district with approximately 50 elementary schools compared with 23 among low-LEP schools. Thus, LEP concentration is correlated with both the demographic composition of the school and the local schooling environment. As a result, I interpret the heterogeneity results as descriptive evidence on where rating responses differ rather than as causal effects of school choice or LEP concentration.

Table 8 reports enrollment responses separately by these school environments and compares Hispanic LEP students with all non-LEP students within each subsample. Panel A shows this comparison across high- and low-choice districts. The hypothesis

is that the enrollment response should be stronger in high-choice districts given that parents have schooling options around them. However, there is little evidence that the Hispanic LEP response is stronger in high-choice districts. At the Acceptable threshold, the estimated enrollment response is 3.4 percent for Hispanic LEP students and not statistically distinguishable from zero. It is even weaker along the Recognized and Exemplary threshold. In comparison, the non-LEP response is much stronger and higher in high-choice districts than the full sample response on the Exemplary margin. However, the Hispanic LEP and non-LEP differences are not statistically distinguishable from zero on any threshold. Meanwhile, the estimates in low-choice districts are considerably less precise, particularly at the Acceptable threshold, and also do not show a clear pattern of differential responsiveness.

The pattern is somewhat more suggestive when schools are divided by LEP concentration. At the Acceptable threshold, Hispanic LEP enrollment increases by 7.2 percent in high-LEP schools and the estimate is statistically significant at the 10 percent level. The corresponding estimate for all non-LEP students is 4.6 percent. In contrast, among low-LEP schools, the Hispanic LEP point estimate at the Acceptable threshold is negative but it is very imprecisely estimated. At the higher thresholds, Hispanic LEP responses remain small in both high- and low-LEP schools. The response for all non-LEP students in comparison follows the same pattern as before. They show a positive 3.7 percent response at the Exemplary threshold in high-LEP schools, which corresponds to a 1.8 percent Hispanic LEP response. Further, none of the within-subsample differences between Hispanic LEP and all non-LEP students are statistically significant. It is worth noting that all the estimates for Hispanic LEP in high-LEP schools are higher in magnitude than the full sample estimates. And given the strong response on the Acceptable threshold, it appears that Hispanic LEP students are responsive of the rating in schools with students similar to their linguistic status.

I also test whether the Hispanic LEP response is different in the high and low environment directly in Table 9. Panel A shows that the differences between high- and low-choice districts are small relative to their standard errors at all three rating thresholds. For example, at the Acceptable threshold, the Hispanic LEP response is 3.4 percent in

high-choice districts but the coefficient in the low-choice districts is too noisy given the small sample size to produce credible differences. The same holds true at the Recognized and Exemplary thresholds. The results therefore provide little evidence that the number of schooling options alone explains the muted average Hispanic LEP response.

Panel B is more interesting given the stronger response in high-LEP schools. However, the large standard errors in the low-LEP sample results in imprecisely estimated differences.

Overall, the heterogeneity analysis does not provide statistically precise evidence that Hispanic LEP responses differ across either the local choice environment or the concentration of LEP students in the school. Nevertheless, the point estimates suggest one potentially interesting pattern in the positive Acceptable-margin response among Hispanic LEP students in high-LEP schools. This pattern is consistent with several possible explanations, including stronger information networks, language-related school resources, or preferences for schools serving larger LEP communities. Since high-LEP schools also differ substantially in urbanicity, economic disadvantage, school size, and the number of nearby schooling options, the estimates cannot distinguish among these mechanisms.

6 Conclusion

School accountability ratings are intended to make information about school quality easier for families to use, but their effectiveness depends on whether families can access and act on these signals. This paper studies how families of Limited English Proficient students respond to such ratings using discontinuities generated by the Texas school accountability system. I find little evidence that Hispanic LEP enrollment responds to crossing any of the three rating thresholds while some non-LEP groups show positive responses at particular margins.

The clearest contrast appears at the Exemplary threshold where enrollment and incoming transfers of all non-LEP students increase compared to the negative and imprecise estimates of Hispanic LEP students. More demographically comparable non-LEP groups also show positive responses at the Acceptable threshold but these estimates

are less precise and more sensitive to bandwidth specification. Therefore, the results point to a muted response among Hispanic LEP families.

The more important question raised by these results is why Hispanic LEP families appear less responsive to the rating signal even when there is some suggestive evidence of other groups responding to it. A weak response could arise because ratings do not reach or become salient to these families, or they face constraints that limit their ability to change schools, or they place little weight on the categorical label. The heterogeneity analysis does not clearly distinguish among these explanations. I find little evidence that a larger local choice set of schools alone generates stronger responses. However, I do find a positive response among Hispanic LEP students in schools with high concentration of LEP students, which suggests that some feature of the school environment may matter. These schools differ along several dimensions and data limitation prevent this study from providing a clean interpretation of the mechanism.

The identification of these mechanisms matter for the effectiveness of the school accountability systems. The publication of school ratings provide an easily interpretable signal on school quality but if families facing greater information or choice frictions are seemingly less responsive to it, these policies may not be benefiting all households equally. So, the accountability systems need to consider not only how to effectively summarize school quality information but also how to reach families and whether all households have meaningful opportunities to act on it.

An important direction of future research for the literature on school accountability policies and LEP families is to investigate the source of this muted response to school ratings. Identifying whether it stems from limited information reach, constraints on school choice, or differences in how families perceive school quality will have different policy implications. The appropriate response could range from improving the delivery of school quality information to reducing barriers to switching schools to providing richer information than categorical ratings based on academic performance alone.

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Tables

Table 1. Descriptive Statistics by Student Group

	Hispanic LEP	All Non-LEP	Hispanic Non-LEP
Panel A. Student and School Characteristics			
Ever Economically Disadvantaged	0.974	0.636	0.848
Urban School	0.614	0.439	0.568
Suburban School	0.269	0.337	0.259
Town/Rural School	0.117	0.224	0.173
Mean Group Enrollment	154.69	400.15	128.04
School-Year Observations	15,553	15,793	15,759
Panel B. Accountability Rating Distribution			
Unacceptable	0.020	0.010	0.009
Acceptable	0.521	0.343	0.413
Recognized	0.386	0.454	0.462
Exemplary	0.073	0.193	0.116

Notes: This table reports descriptive statistics for Hispanic LEP, all non-LEP, and Hispanic non-LEP students in Texas elementary schools from 2001 to 2008, excluding 2003. Panel A reports student-weighted school characteristics, except for mean group enrollment, which is the unweighted mean number of students in the corresponding group at the school-year level. School-year observations report the number of observations with positive enrollment for the corresponding group. Panel B reports the student-weighted distribution of each group's enrollment across school accountability rating categories.

Table 2. Balance on Baseline Characteristics for LEP Students

	Acceptable	Recognized	Exemplary
Total LEP Enrollment (t)	3.68 (40.230)	0.64 (11.402)	12.87 (8.500)
Mean	241.27	181.03	115.16
Hispanic LEP Enrollment (t)	5.01 (39.356)	-2.98 (11.201)	13.35 (8.406)
Mean	231.63	167.40	98.01
Hispanic LEP Enrollment Change Ratio ($t - 2$ to t)	-0.05 (0.084)	-0.02 (0.059)	-0.16 (0.171)
Mean	0.25	0.24	0.38
Hispanic LEP Leaving Ratio (t & $t - 2$)	0.00 (0.028)	0.01 (0.011)	0.03** (0.014)
Mean	0.26	0.25	0.27
Hisp. LEP Inc. Trans. Ratio (t & $t - 2$)	-0.01 (0.056)	-0.02 (0.040)	-0.05 (0.078)
Mean	0.40	0.43	0.52
Hisp. LEP New Entrant Ratio (t & $t - 2$)	-0.02 (0.050)	0.00 (0.030)	-0.12 (0.116)
Mean	0.40	0.40	0.50

Notes: Standard errors, clustered at the school level, are reported in parentheses. Estimates are calculated within a bandwidth of 8. All variables are measured at baseline. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 3. Balance on Baseline Characteristics for Non-LEP Students

	Acceptable	Recognized	Exemplary
Total Non-LEP Enrollment (t)	-23.50 (27.619)	17.81* (10.821)	-12.03 (12.265)
Mean	296.68	389.38	446.85
Hispanic Non-LEP Enrollment (t)	-9.13 (16.277)	6.20 (7.351)	1.69 (6.887)
Mean	102.66	140.92	114.08
All Non-LEP Enrollment Change Ratio ($t - 2$ to t)	0.02 (0.038)	0.02** (0.009)	-0.01 (0.010)
Mean	-0.03	-0.01	0.00
Hispanic Non-LEP Enrollment Change Ratio ($t - 2$ to t)	-0.01 (0.066)	0.03 (0.022)	0.01 (0.025)
Mean	0.08	0.10	0.13
All Non-LEP Leaving Ratio (t & $t - 2$)	0.04 (0.030)	0.01 (0.008)	0.01 (0.007)
Mean	0.36	0.29	0.24
All Non-LEP Inc. Trans. Ratio (t & $t - 2$)	0.06* (0.038)	0.01 (0.008)	0.00 (0.008)
Mean	0.36	0.31	0.26
All Non-LEP New Entrant Ratio (t & $t - 2$)	0.03 (0.035)	0.01* (0.008)	-0.01 (0.009)
Mean	0.33	0.34	0.36

Notes: Standard errors, clustered at the school level, are reported in parentheses. Estimates are calculated within a bandwidth of 8. All variables are measured at baseline. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 4. Effect of Crossing a Rating Threshold on Enrollment Change, by Student Group

	Hispanic LEP	All Non-LEP	Hispanic Non-LEP	Economically Disadvantaged Non-LEP
Acceptable	0.027 (0.058)	0.054 (0.036)	0.165* (0.088)	0.043 (0.034)
Outcome Mean	0.06	-0.02	0.16	-0.01
Recognized	-0.019 (0.040)	0.005 (0.009)	-0.037 (0.024)	0.006 (0.009)
Outcome Mean	0.14	-0.01	0.08	-0.00
Exemplary	-0.031 (0.049)	0.019** (0.008)	0.023 (0.023)	0.007 (0.012)
Outcome Mean	0.22	-0.01	0.10	0.00

Notes: Standard errors, clustered at the school level, are reported in parentheses. RD estimates use a bandwidth of 8 and are estimated separately for each student group. The sample includes observations from 2001–2008, excluding 2003. The outcome is the proportional change in enrollment between year t and year $t + 2$, normalized by the corresponding group’s enrollment in year t . The numbers of observations at the Acceptable, Recognized, and Exemplary thresholds are 776, 5,737, and 5,438 for Hispanic LEP students; 756, 5,642, and 5,430 for all non-LEP students; 784, 5,811, and 5,540 for Hispanic non-LEP students; and 787, 5,818, and 5,548 for economically disadvantaged non-LEP students.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 5. Comparison of Enrollment Responses Between Hispanic LEP Students and Non-LEP Groups

	All Non-LEP	Hispanic Non-LEP	Economically Disadvantaged Non-LEP
Acceptable			
Comparison Group	0.054 (0.037)	0.165** (0.082)	0.043 (0.034)
Hispanic LEP	0.027 (0.055)	0.027 (0.055)	0.027 (0.055)
Difference	-0.027 (0.065)	-0.138 (0.089)	-0.016 (0.063)
Recognized			
Comparison Group	0.005 (0.008)	-0.037 (0.024)	0.006 (0.009)
Hispanic LEP	-0.019 (0.039)	-0.019 (0.039)	-0.019 (0.039)
Difference	-0.024 (0.040)	0.018 (0.045)	-0.025 (0.039)
Exemplary			
Comparison Group	0.019** (0.007)	0.023 (0.023)	0.007 (0.012)
Hispanic LEP	-0.031 (0.046)	-0.031 (0.046)	-0.031 (0.046)
Difference	-0.050 (0.047)	-0.055 (0.050)	-0.038 (0.047)

Notes: Each column reports estimates from a separate specification comparing Hispanic LEP students with the comparison group indicated in the column heading. The “Comparison Group” and “Hispanic LEP” rows report the estimated discontinuity in enrollment change for each group. The “Difference” row reports the coefficient on the interaction between Hispanic LEP status and crossing the relevant rating threshold, and therefore measures the Hispanic LEP estimate minus the corresponding comparison-group estimate. Negative differences indicate a smaller enrollment response among Hispanic LEP students. Standard errors, clustered at the school level, are reported in parentheses. RD estimates use a bandwidth of 8. The sample includes observations from 2001–2008, excluding 2003. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 6. Effect of Crossing a Rating Threshold on Enrollment Channels, by Student Group

	Hispanic LEP	All Non-LEP	Hispanic Non-LEP	Economically Disadvantaged Non-LEP
Panel A: Incoming Transfer Ratio				
Acceptable	0.033 (0.050)	0.095*** (0.029)	0.177* (0.107)	0.103*** (0.031)
Recognized	-0.023 (0.026)	0.003 (0.008)	-0.019 (0.019)	0.002 (0.009)
Exemplary	-0.034 (0.036)	0.015** (0.007)	0.006 (0.015)	-0.003 (0.011)
Panel B: New Entrant Ratio				
Acceptable	0.002 (0.036)	0.016 (0.033)	0.075 (0.055)	0.011 (0.032)
Recognized	0.005 (0.019)	0.005 (0.007)	-0.015 (0.013)	0.003 (0.008)
Exemplary	0.007 (0.025)	0.007 (0.006)	0.012 (0.014)	0.003 (0.008)
Panel C: Leaver Ratio				
Acceptable	0.012 (0.028)	0.041 (0.031)	0.025 (0.035)	0.042 (0.031)
Recognized	0.007 (0.009)	0.005 (0.007)	0.003 (0.009)	0.001 (0.008)
Exemplary	0.008 (0.012)	0.001 (0.005)	-0.002 (0.007)	-0.002 (0.006)

Notes: Standard errors, clustered at the school level, are reported in parentheses. RD estimates use a bandwidth of 8. The specifications are unweighted. The sample includes observations from 2001–2008, excluding 2003. Incoming transfers, new entrants, and leavers are normalized by the corresponding group's enrollment in year t . * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 7. Descriptive Characteristics by School Environment

	Low	High	Difference
Panel A: District Choice			
Total School Enrollment	342.339	611.636	269.296
LEP Share	0.106	0.288	0.182
Economically Disadvantaged Share	0.547	0.509	-0.038
Hispanic LEP Share	0.102	0.264	0.162
Urban School	0.042	0.525	0.483
Suburban School	0.086	0.318	0.231
Town/Rural School	0.871	0.157	-0.714
Number of Elementary Schools in District	1.000	42.622	41.622
Panel B: LEP Concentration			
Total School Enrollment	516.234	634.440	118.206
LEP Share	0.072	0.453	0.381
Economically Disadvantaged Share	0.441	0.589	0.148
Hispanic LEP Share	0.057	0.426	0.370
Urban School	0.332	0.579	0.247
Suburban School	0.292	0.277	-0.015
Town/Rural School	0.375	0.143	-0.232
Number of Elementary Schools in District	23.295	50.274	26.979

Notes: The table reports unweighted school-year means by school environment. Panel A compares low- and high-choice districts, where high-choice districts are those with a number of elementary schools above the district-level median. Panel B compares schools with LEP enrollment shares below or at versus above the median LEP share. The “Difference” column reports the high-group mean minus the low-group mean. The sample includes Texas elementary schools from 2001–2008, excluding 2003.

Table 8. Heterogeneity in Enrollment Responses by School Environment

	Acceptable	Recognized	Exemplary
Panel A: District Choice			
<i>High-Choice Districts</i>			
All Non-LEP	0.037 (0.039)	0.006 (0.010)	0.026*** (0.008)
Hispanic LEP	0.034 (0.047)	-0.029 (0.043)	-0.039 (0.051)
Difference	-0.002 (0.058)	-0.035 (0.044)	-0.065 (0.051)
<i>Low-Choice Districts</i>			
All Non-LEP	0.106 (0.101)	-0.004 (0.013)	-0.018 (0.014)
Hispanic LEP	0.017 (0.262)	0.037 (0.081)	0.021 (0.113)
Difference	-0.089 (0.278)	0.042 (0.081)	0.039 (0.115)
Panel B: LEP Concentration			
<i>High-LEP Schools</i>			
All Non-LEP	0.046 (0.044)	0.001 (0.013)	0.037** (0.015)
Hispanic LEP	0.072* (0.039)	-0.006 (0.015)	0.018 (0.024)
Difference	0.026 (0.054)	-0.007 (0.019)	-0.019 (0.028)
<i>Low-LEP Schools</i>			
All Non-LEP	-0.018 (0.046)	0.013 (0.009)	0.006 (0.008)
Hispanic LEP	-0.050 (0.180)	-0.039 (0.088)	-0.038 (0.069)
Difference	-0.032 (0.181)	-0.052 (0.088)	-0.045 (0.070)

Notes: The table reports RD estimates for the enrollment change ratio separately by school environment. Panel A splits the sample into high- and low-choice districts based on the number of elementary schools in the district. Panel B splits the sample by LEP concentration. The “Difference” row reports the Hispanic LEP estimate minus the all non-LEP estimate within the corresponding subsample. Standard errors, clustered at the school level, are reported in parentheses. RD estimates use a bandwidth of 8. The specifications are unweighted. The sample includes observations from 2001–2008, excluding 2003. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 9. Heterogeneity in Enrollment Responses Among Hispanic LEP Students

	Acceptable	Recognized	Exemplary
Panel A: District Choice			
High-Choice	0.034 (0.047)	-0.029 (0.043)	-0.039 (0.051)
Low-Choice	0.017 (0.259)	0.037 (0.080)	0.021 (0.113)
Difference	0.018 (0.263)	-0.066 (0.091)	-0.060 (0.124)
Panel B: LEP Concentration			
High-LEP	0.072* (0.039)	-0.006 (0.015)	0.018 (0.024)
Low-LEP	-0.050 (0.178)	-0.039 (0.088)	-0.038 (0.069)
Difference	0.122 (0.182)	0.033 (0.089)	0.056 (0.073)

Notes: The table reports heterogeneity in the RD effect on the enrollment change ratio among Hispanic LEP students. Panel A compares Hispanic LEP students in high- and low-choice districts, where district choice is measured using the number of elementary schools in the district. Panel B compares Hispanic LEP students in high- and low-LEP schools. The “Difference” row reports the estimated difference between the high and low subsamples (High – Low). Standard errors, clustered at the school level, are reported in parentheses. RD estimates use a bandwidth of 8. The specifications are unweighted. The sample includes observations from 2001–2008, excluding 2003. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Figures

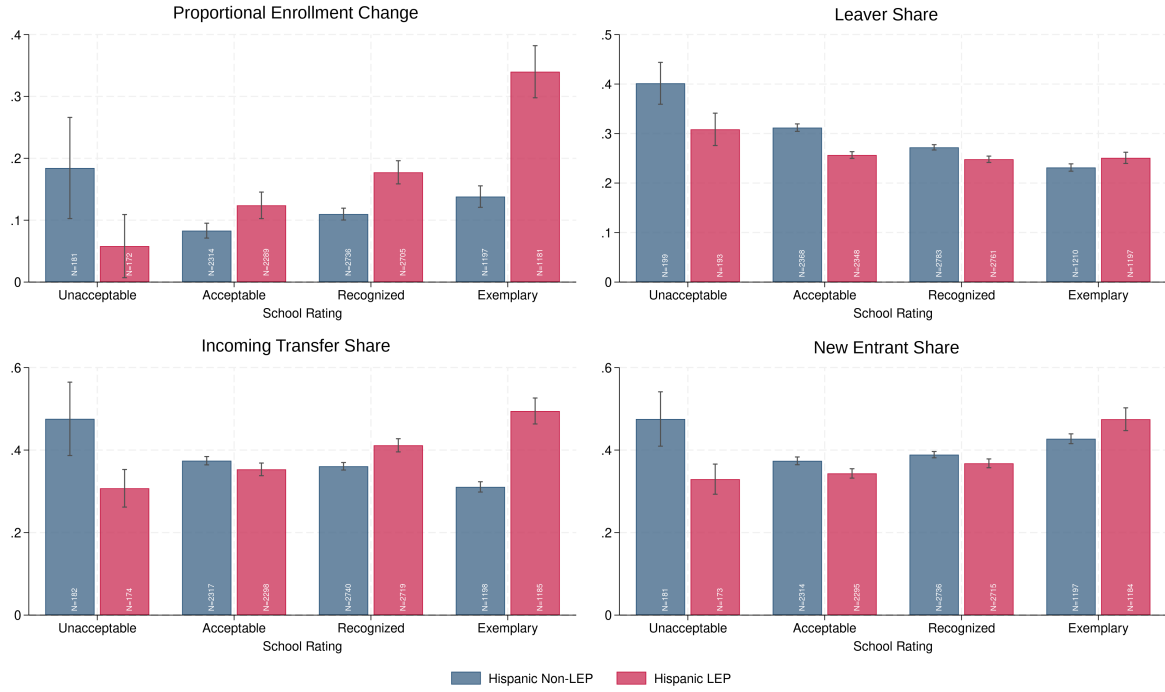


Figure 1. Raw Means by Rating Category and LEP Status

Notes: The figure reports raw mean enrollment outcomes for Hispanic LEP and Hispanic non-LEP students by school accountability rating. Outcomes are measured between the rating year, t , and year $t + 2$ and are normalized by the corresponding group's enrollment in year t . Enrollment change is the proportional change in group enrollment between t and $t + 2$. The remaining panels report the shares of year- t enrollment represented by leavers, nonstructural incoming transfers, and new entrants. Leavers exclude students expected to leave because of grade progression, and incoming transfers exclude structural feeder-school moves. Error bars report 95 percent confidence intervals. These are unconditional descriptive means and should not be interpreted as causal effects of school ratings.

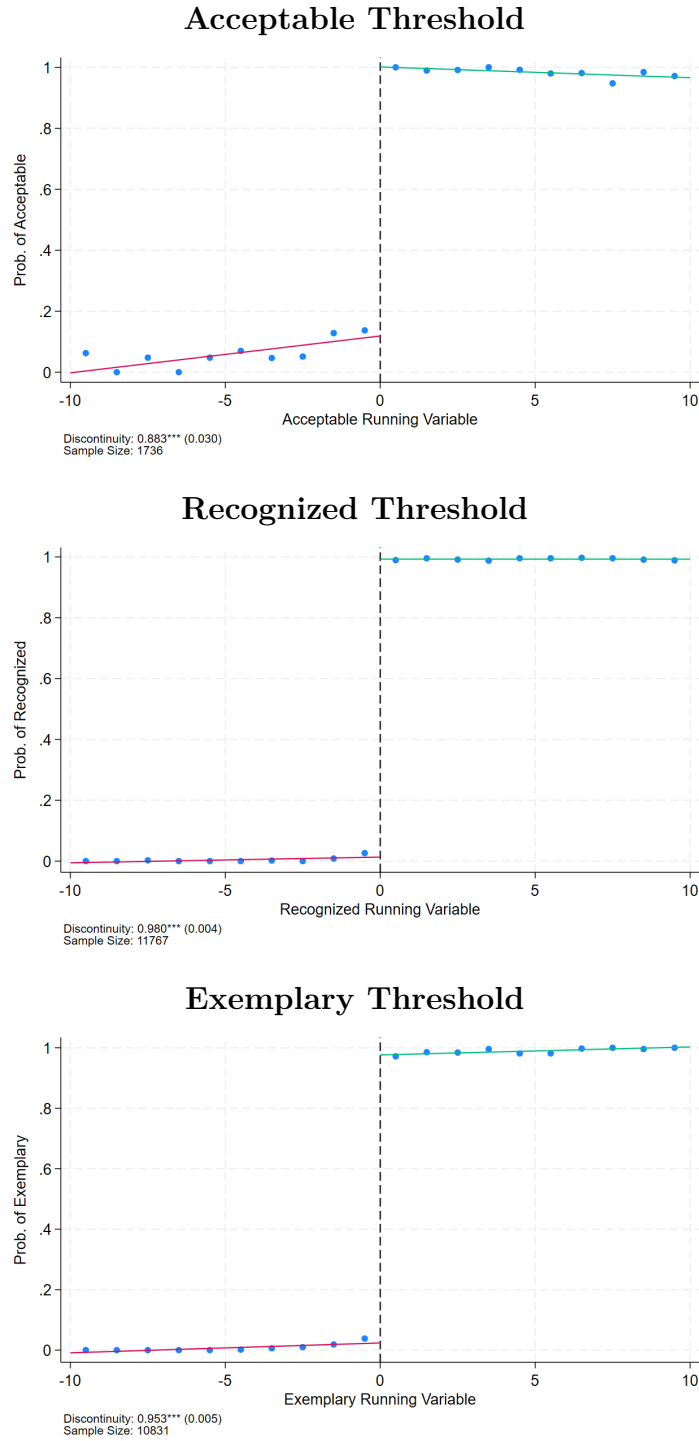


Figure 2. Probability a School Attains a Higher Rating, by Rating Category

Notes: The figure plots the probability that a school receives the higher accountability rating against the running variable at each rating threshold. The running variable is the pass rate of the binding subject–demographic group relative to the applicable statutory cutoff and is normalized to zero at the threshold. Dots report mean rating attainment within bins of the running variable, and solid lines show separate linear fits on either side of the cutoff. The dashed vertical line denotes the rating threshold.

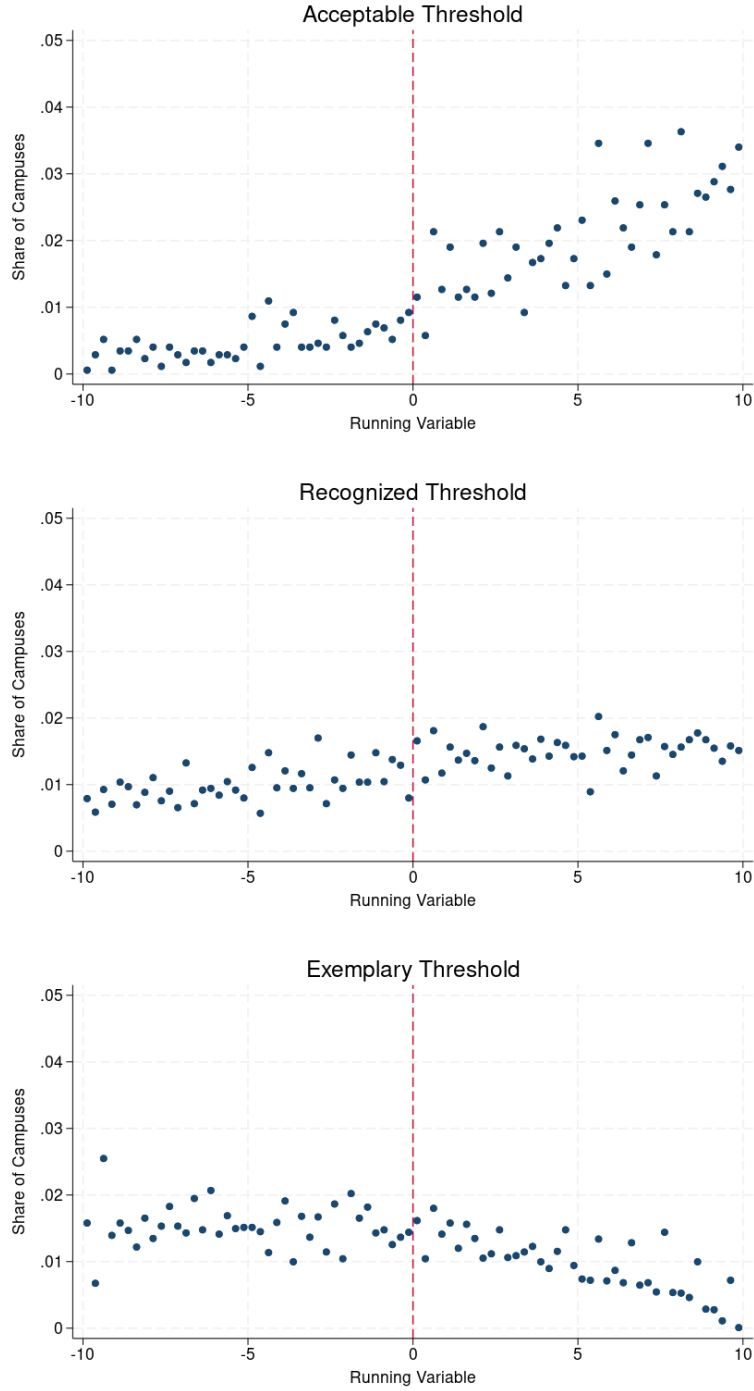


Figure 3. Running variable density, by accountability rating threshold

Notes: The figure plots the distribution of the running variable separately for the Acceptable, Recognized, and Exemplary rating thresholds. The running variable is defined as the pass rate of the binding subject–demographic group minus the applicable statutory cutoff and is therefore centered at zero. Each dot reports the share of campus-year observations falling within a 0.25 percentage-point bin. The dashed vertical line denotes the rating threshold.

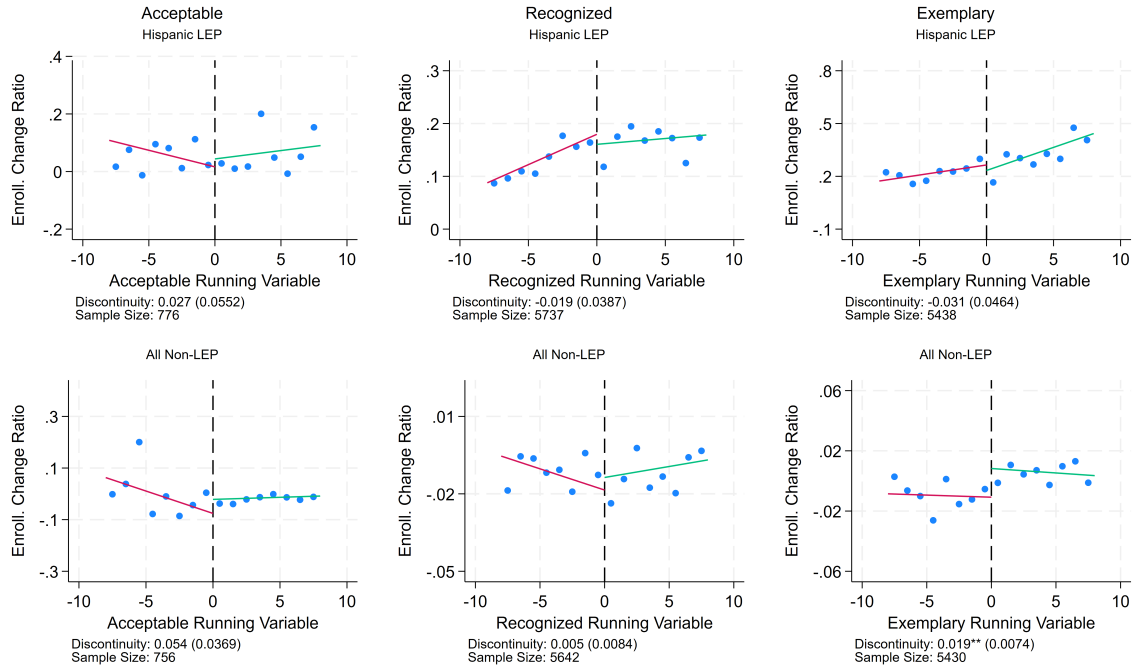


Figure 4. Enrollment Change Around Accountability Rating Thresholds

Notes: The figure plots the enrollment change ratio against the centered running variable at the Acceptable, Recognized, and Exemplary rating thresholds. The top row reports Hispanic LEP students and the bottom row reports all non-LEP students. Dots show mean outcomes within bins of the running variable, and solid lines show separate linear fits on either side of the cutoff. The dashed vertical line denotes the statutory rating threshold. The enrollment change ratio measures the proportional change in group enrollment between year t and year $t + 2$, normalized by group enrollment in year t . Each panel reports the estimated discontinuity at the cutoff, with the school-clustered standard error in parentheses, and the number of school-year observations. RD estimates use a bandwidth of 8. The sample covers 2001–2008, excluding 2003.

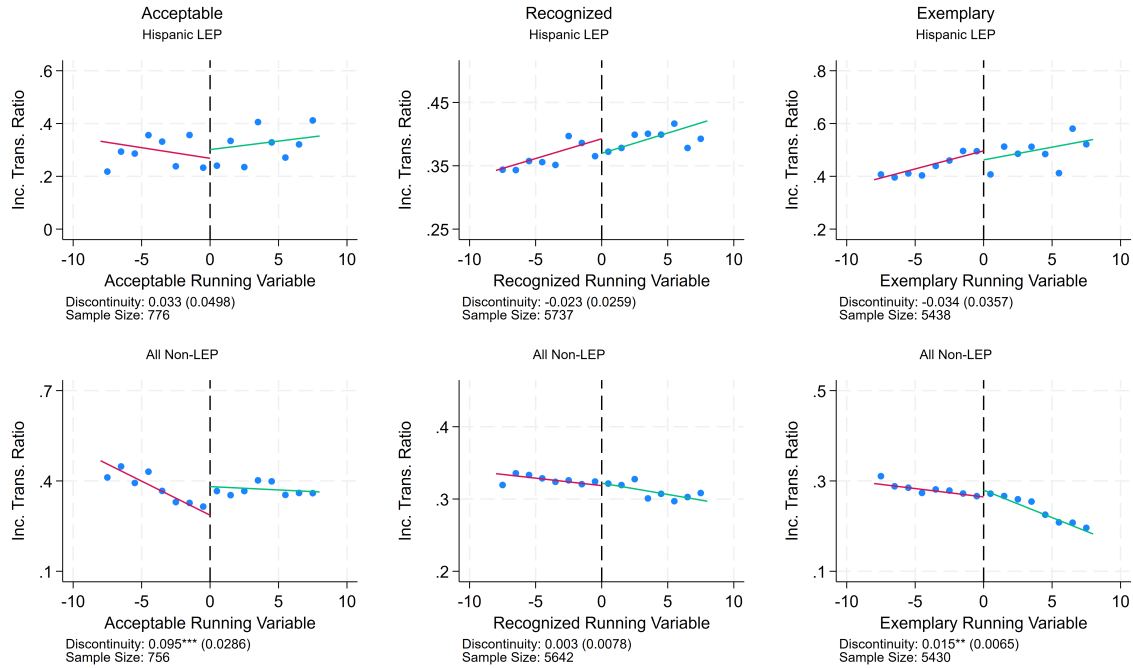


Figure 5. Incoming Transfer Around Accountability Rating Thresholds

Notes: The figure plots the enrollment change ratio against the centered running variable at the Acceptable, Recognized, and Exemplary rating thresholds. The top row reports Hispanic LEP students and the bottom row reports all non-LEP students. Dots show mean outcomes within bins of the running variable, and solid lines show separate linear fits on either side of the cutoff. The dashed vertical line denotes the statutory rating threshold. The incoming transfer ratio is the number of nonstructural incoming transfers between year t and year $t + 2$, normalized by group enrollment in year t . Each panel reports the estimated discontinuity at the cutoff, with the school-clustered standard error in parentheses, and the number of school-year observations. RD estimates use a bandwidth of 8. The sample covers 2001–2008, excluding 2003.

A Appendix

A.1 Tables

Table A1. Balance on Baseline Characteristics for Non-LEP Students: Alternative Bandwidth

	Acceptable	Recognized	Exemplary
Total Non-LEP Enrollment (t)	-38.50 (38.007)	27.70* (14.392)	-16.97 (15.953)
Mean	293.12	387.60	442.19
Hispanic Non-LEP Enrollment (t)	4.13 (22.065)	0.08 (9.806)	-1.77 (8.641)
Mean	100.17	141.09	116.86
All Non-LEP Enrollment Change Ratio ($t - 2$ to t)	-0.03 (0.053)	0.02* (0.013)	-0.02 (0.014)
Mean	-0.03	-0.01	-0.00
Hispanic Non-LEP Enrollment Change Ratio ($t - 2$ to t)	-0.10 (0.091)	0.04 (0.028)	0.00 (0.032)
Mean	0.06	0.09	0.13
All Non-LEP Leaving Ratio (t & $t - 2$)	0.05 (0.039)	0.00 (0.011)	0.01 (0.010)
Mean	0.35	0.30	0.25
All Non-LEP Inc. Trans. Ratio (t & $t - 2$)	0.00 (0.049)	0.02 (0.011)	0.00 (0.011)
Mean	0.36	0.32	0.27
All Non-LEP New Entrant Ratio (t & $t - 2$)	0.01 (0.048)	0.01 (0.011)	-0.02 (0.011)
Mean	0.32	0.34	0.36

Notes: Standard errors, clustered at the school level, are reported in parentheses. Estimates are calculated within a bandwidth of 4. All variables are measured at baseline. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A2. Robustness of Enrollment Effects to Alternative Bandwidths

	Fixed Bandwidth				Optimal Bandwidth
	4	6	8 <i>Baseline</i>	10	Optimal
Panel A: Hispanic LEP					
Acceptable	-0.015 (0.056)	0.069 (0.065)	0.027 (0.058)	0.015 (0.053)	-0.010 (0.060) [3.50]
Recognized	-0.006 (0.053)	-0.006 (0.045)	-0.019 (0.040)	-0.029 (0.036)	-0.002 (0.043) [6.10]
Exemplary	-0.019 (0.063)	-0.053 (0.056)	-0.031 (0.049)	-0.027 (0.045)	-0.039 (0.074) [2.23]
Panel B: All Non-LEP					
Acceptable	0.009 (0.045)	0.008 (0.037)	0.054 (0.036)	0.013 (0.030)	0.031 (0.040) [6.91]
Recognized	0.016 (0.012)	0.012 (0.010)	0.005 (0.009)	0.002 (0.008)	0.010 (0.010) [5.57]
Exemplary	0.019* (0.010)	0.015* (0.008)	0.019** (0.008)	0.020*** (0.007)	0.020 (0.013) [2.40]

Table A2. Robustness of Enrollment Effects to Alternative Bandwidths (continued)

	Fixed Bandwidth				Optimal Bandwidth
	4	6	8 <i>Baseline</i>	10	Optimal
Panel C: Hispanic Non-LEP					
Acceptable	0.069 (0.098)	0.106 (0.091)	0.165* (0.088)	0.102 (0.070)	0.069 (0.098) [4.05]
Recognized	-0.023 (0.033)	-0.021 (0.028)	-0.037 (0.024)	-0.034 (0.022)	-0.037 (0.024) [8.02]
Exemplary	0.040 (0.028)	0.015 (0.025)	0.023 (0.023)	0.019 (0.021)	0.037 (0.029) [2.91]
Panel D: Economically Disadvantaged Non-LEP					
Acceptable	-0.005 (0.043)	-0.003 (0.035)	0.043 (0.034)	0.006 (0.028)	0.021 (0.038) [6.77]
Recognized	0.021 (0.013)	0.016 (0.010)	0.006 (0.009)	0.003 (0.008)	0.020 (0.012) [4.15]
Exemplary	0.021 (0.013)	0.012 (0.011)	0.007 (0.012)	0.016 (0.010)	0.018 (0.018) [2.20]

Notes: The table reports RD estimates of the effect of crossing each accountability rating threshold on the enrollment change ratio using alternative bandwidths. Columns (1)–(4) use fixed bandwidths of 4, 6, 8, and 10, respectively. A bandwidth of 8 is used in the baseline specification reported in Table 4. The final column uses the optimal bandwidth selected using the bandwidth selectors for local polynomial regression discontinuity point estimation and inference developed by Calonico et al. (2014). The selected bandwidth is reported in square brackets. Standard errors, clustered at the school level, are reported in parentheses. Specifications are unweighted and use the same controls and fixed effects as the baseline specification. The sample includes observations from 2001–2008, excluding 2003. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A3. Robustness of Incoming Transfer Effects to Alternative Bandwidths

	Fixed Bandwidth				Optimal Bandwidth
	4	6	8 <i>Baseline</i>	10	Optimal
Panel A: Hispanic LEP					
Acceptable	0.027 (0.058)	0.089 (0.057)	0.033 (0.051)	0.051 (0.047)	0.048 (0.059) [3.47]
Recognized	-0.021 (0.034)	-0.020 (0.029)	-0.023 (0.027)	-0.030 (0.024)	-0.022 (0.026) [7.05]
Exemplary	-0.018 (0.049)	-0.045 (0.044)	-0.034 (0.038)	-0.016 (0.034)	-0.012 (0.056) [2.37]
Panel B: All Non-LEP					
Acceptable	0.056 (0.036)	0.078*** (0.029)	0.095*** (0.029)	0.073*** (0.025)	0.043 (0.038) [3.66]
Recognized	0.011 (0.011)	0.008 (0.009)	0.003 (0.008)	0.002 (0.008)	0.007 (0.009) [6.03]
Exemplary	0.012 (0.010)	0.014* (0.008)	0.015** (0.008)	0.022*** (0.007)	0.006 (0.013) [2.35]

Table A3. Robustness of Incoming Transfer Effects to Alternative Bandwidths (continued)

	Fixed Bandwidth				Optimal Bandwidth
	4	6	8 <i>Baseline</i>	10	Optimal
Panel C: Hispanic Non-LEP					
Acceptable	0.080 (0.098)	0.128 (0.082)	0.177* (0.107)	0.131* (0.068)	0.073 (0.100) [3.86]
Recognized	-0.020 (0.026)	-0.017 (0.022)	-0.019 (0.020)	-0.016 (0.017)	-0.017 (0.019) [8.16]
Exemplary	0.022 (0.019)	0.016 (0.017)	0.006 (0.015)	0.013 (0.014)	0.006 (0.024) [1.84]
Panel D: Economically Disadvantaged Non-LEP					
Acceptable	0.058 (0.041)	0.088*** (0.032)	0.103*** (0.031)	0.073*** (0.026)	0.083** (0.034) [5.73]
Recognized	0.016 (0.012)	0.009 (0.010)	0.002 (0.009)	0.003 (0.008)	0.010 (0.010) [5.86]
Exemplary	0.009 (0.013)	0.003 (0.011)	-0.003 (0.012)	0.007 (0.010)	-0.001 (0.019) [1.79]

Notes: The table reports RD estimates of the effect of crossing each accountability rating threshold on the incoming transfer ratio using alternative bandwidths. Columns (1)–(4) use fixed bandwidths of 4, 6, 8, and 10, respectively. A bandwidth of 8 is used in the baseline specification reported in Table 6. The final column uses the optimal bandwidth selected using the bandwidth selectors for local polynomial regression discontinuity point estimation and inference developed by Calonico et al. (2014). The selected bandwidth is reported in square brackets. Incoming transfers are normalized by the corresponding group’s enrollment in year t . Standard errors, clustered at the school level, are reported in parentheses. Specifications are unweighted and use the same controls and fixed effects as the baseline specification. The sample includes observations from 2001–2008, excluding 2003. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

A.2 Figures

TEXAS EDUCATION AGENCY

Page 1

School Name: STAFFORD ELEMENTARY SCHOOL

2004-05 School Report Card

School Enrollment: 568

School Number: 015905114

Grade Span: PK - 05

2005 School Accountability Rating: Recognized

School Type: Elementary

Gold Performance Acknowledgments:

Commended on Reading/ELA

Commended on Mathematics

Commended on Science

Comparable Improvement: Mathematics

District Name: EDGEWOOD ISD

2005 District Accountability Rating: Academically Acceptable

The complete report for this school is the 2004-05 Academic Excellence Indicator System (AEIS) Report

Questions? See the School Report Card Definitions in English, Definiciones in Spanish.

		State Average	District Average	School Group Median	School (All Students)	African American	Hispanic	White	Native American	Asian/ Pac.Is.	Econ. Disadv.
<u>TAKS Met 2005 Passing Std (Sum of All Grades Tested)</u>											
Reading/ELA	2005	83%	71%	76%	82%	80%	82%	83%	+	+	81%
	2004	80%	68%	71%	75%	> 99%	75%	50%	+	+	75%
Mathematics	2005	72%	55%	73%	86%	80%	87%	80%	+	+	86%
	2004	67%	50%	72%	72%	83%	73%	50%	+	+	72%
Writing	2005	90%	84%	88%	98%	+	98%	+	+	+	98%
	2004	89%	86%	88%	84%	+	85%	+	+	+	84%
Science	2005	66%	47%	46%	74%	+	74%	+	+	+	74%
	2004	60%	36%	44%	48%	+	47%	+	+	+	48%
All Tests	2005	62%	44%	56%	71%	80%	71%	67%	+	+	70%
	2004	58%	38%	56%	57%	83%	57%	33%	+	+	56%
<u>SDAA II Examinations (Sum of Grades 3-10)</u>											
<u>Met ABD Expectations</u>											
	2005	79%	70%	90%	73%	+	72%	+	+	+	73%
<u>TAKS/SDAA II Exemptions</u>											
ARD Exempt	2005	0.8%	1.3%	0.0%	0.5%	0.0%	0.5%	0.0%	-	-	0.5%
	2004	2.1%	3.8%	0.0%	0.8%	0.0%	0.8%	0.0%	-	-	0.8%
LEP Exempt	2005	1.0%	0.8%	1.7%	0.0%	0.0%	0.0%	0.0%	-	-	0.0%
	2004	1.2%	0.8%	1.4%	0.8%	0.0%	0.8%	0.0%	-	-	0.8%
<u>Progress of Prior Year TAKS Failures (Sum of Gr 4-11)</u>											
<u>Percent of Failures Passing TAKS</u>											
Reading/ELA	2005	45%	31%	48%	44%	+	44%	+	+	+	44%
	2004	47%	40%	31%	36%	+	38%	+	+	+	36%
Mathematics	2005	25%	15%	38%	36%	+	33%	+	+	+	36%
	2004	27%	22%	33%	38%	+	38%	+	+	+	38%
<u>Student Success Initiative</u>											
<u>Grade 3 Reading (English and Spanish)</u>											
<u>Students Requiring Accelerated Instruction</u>											
	2005	13%	20%	23%	12%	+	11%	+	+	+	13%
	2004	10%	19%	15%	8%	+	7%	+	+	+	9%
<u>TAKS Cumulative Met Standard (1st & 2nd Admin)</u>											
	2005	93%	87%	88%	91%	+	91%	+	+	+	91%
	2004	95%	89%	92%	97%	+	98%	+	+	+	96%
<u>Grade 5 Reading (English and Spanish)</u>											
<u>Students Requiring Accelerated Instruction</u>											
	2005	25%	36%	44%	27%	+	26%	+	+	+	27%
<u>TAKS Cumulative Met Standard (1st & 2nd Admin)</u>											
	2005	86%	77%	72%	83%	+	81%	+	+	+	83%

Figure A1. Example of a Texas School Report Card, 2004–05

Notes: The figure shows an excerpt from a 2004–05 Texas School Report Card. The school's accountability rating appears near the top of the report alongside more detailed measures of student performance.